

Code of Practice

GM/RCI500

Issue One

Date December 1994

Code of Practice for EMC Between the Railway and its Neighbourhood

Approved by

Signatures removed from electronic version

B J Wojtas

Standards Manager, Trains & Train Systems Unit

Authorised by

J R Mitchell

Chairman, Traction and Rolling Stock Subject Committee

Synopsis

This code of Practice defines the electromagnetic compatibility at the interfaces between railways and the outside world, including the public spaces within the railways

This document is the property of Railtrack. It shall not be reproduced in whole or in part without the written permission of the Controller, Safety Standards.

Published by

**Safety & Standards Directorate
Railtrack
Railway Technical Centre
London Road
Derby**

©Copyright 1994 Railtrack

This page is intentionally blank.

Code of Practice for EMC Between the Railways and its Neighbourhood

Code of Practice

GM/RC1500

Issue One

Date December 1994

Page 1 of 9

Contents

Section	Description	Page
Part A	Issue record	2
	Distribution	2
	Disclaimer	2
	Supply	2
Part B		
1	Purpose	3
2	Scope	3
3	Definitions	3
4	Requirements	4
5	Guidance Notes	8
References		9

Code of Practice

GM/RC I 500

Issue One

Date December 1994

Page 2 of 9

Code of Practice for EMC Between the Railways and its Neighbourhood

Part A

Issue record

This Code of Practice will be updated when necessary by distribution of a complete replacement.

Amended or additional parts of revised pages will be marked by a vertical black line in the adjacent margin.

Issue	Date	Comments
I	Dec 94	Original Document

Distribution

Controlled copies of this Code of Practice should be made available to all people who are responsible for managing EMC.

Disclaimer

The Safety & Standards Directorate (S&SD) shall use its best endeavours to ensure that the content, layout and text of its documents are accurate, complete and in line with current best practice insofar as is reasonably practicable. It makes no warranties, express or implied, that compliance with all or any of its documents shall be sufficient to ensure safe systems of work or operation. The S&SD will not be liable to pay compensation in respect to the content or subsequent use of its standards, except where it can be shown to have acted in bad faith or there has been wilful default.

Supply

Controlled and uncontrolled copies of this code of practice must be obtained from The Catalogue Secretary, Railtrack Safety & Standards Directorate, Floor 2, Fitzroy House, 355 Euston Road, London, NW1 3AG.

Code of Practice for EMC Between the Railways and its Neighbourhood

Code of Practice

GM/RC 1500

Issue One

Date December 1994

Page 3 of 9

Part B

1 Purpose *To assist the total railway with ensuring that it does not generate excessive electromagnetic interference outside its boundary and is tolerant to a reasonable electromagnetic interference from outside its boundary in compliance with the requirements of the EU Electromagnetic Compatibility (EMC) Directive.*

2 Scope *This Code of Practice identifies the limits for railway electromagnetic emission and immunity at the total railway boundary.*

Individual equipment, e.g. S&T apparatus, locomotives, etc., have lower limits of emission to be defined in associated documents.

Although the limits are for electrified lines, the limits identified for 25kV electrified lines also apply to non electrified lines.

Note: Some explanatory notes are added for clarification and these appear as numbered Notes.

3 Definitions

3.1 Total Railway

The assembly of constituent parts which together form an operating railway. These parts include: traction and rolling stock which may operate as trains; the plant, the signalling, telecommunications and control equipment; on electrified lines - the electrical supply distributed along the track which feeds electrical energy to moving vehicles, etc.

3.2 Train Operator (hereafter called Operator)

The body which operates trains.

3.3 Infrastructure

Fixed equipment for which the Infrastructure Controller is responsible which comprises the following:-

- *Power System, comprising power supply, including connection to the supply industry, the conductors along the railway track, the rails, etc,*
- *Signalling System, comprising signalling, including track circuits, cables, interlocking control rooms, etc,*
- *Telecommunication System, comprising telecommunications, including cables, phones, radio links, data communications, etc,*
- *Permanent Way System.*

Code of Practice

GM/RC I 500

Issue One

Date December 1994

Page 4 of 9

Code of Practice for EMC Between the Railways and its Neighbourhood

3.4 Infrastructure Controller (hereafter called Controller)

The person or body that holds legal responsibility in respect of railway network infrastructure.

3.5 Usual Operating Conditions

These include the following:-

- operation under emergency feeding conditions where currents may be greater than usual,
- the passage of trains across gaps in the Power System such as at neutral sections and conductor rail gaps,
- clearance of electrical faults,
- system energisation without movement of traction units,
- raising and lowering pantographs and 3rd rail collector shoes,
- transient behaviour of T&RS such as inrush current into ac transformers and into dc filters,
- the production of transients and surges due to switching actions on the supply, on-train loads and/or the feed from the Electricity Supply Industry,
- the electrical resonant behaviour of ac Power Systems,
- such partial failure or loss of connections between parts of a System as shall be agreed with the Operator.
- the effects of the range of weather conditions appropriate for the location,
- the conditions which may be expected due to the normal operation of the railway.

3.6 Neighbourhood

The physical space external to but adjoining the boundary planes defined in 4.2.2.

4 EMC Requirements

4.1

There are two related aspects of the requirements of EMC, the first concerns the emissions which the installation produces, the second concerns the immunity to outside interference which has to be built into the installation.

Code of Practice for EMC Between the Railways and its Neighbourhood

Code of Practice

GM/RC 1500

Issue One

Date December 1994

Page 5 of 9

4.2 Emission

4.2.1

The emission is identified at arbitrary boundary planes, defined in 4.2.2, which are assumed to separate the railway from the neighbourhood.

4.2.2

The boundary planes are defined as:-

- the two vertical planes which are 10 m laterally beyond the centre lines of the outer railway tracks. If railway apparatus is placed beyond this distance, such as at a substation, the 10 m shall be measured laterally from the outer edges of the apparatus.
- the horizontal plane which is 10 m below the level of the top of the running rails.
- the horizontal plane which is 10 m above the level of the highest current - carrying conductor of the railway or the running rails on non electrified lines. On lines which are likely to be electrified the 10 m dimension above the running rails becomes 15 m above the running rails.

4.2.3

The Controller should ensure that under usual operating conditions, magnetic fields produced by the railway and measured at the boundary planes defined in 4.2.2, do not exceed values given in Table 1.

Table 1 - Permitted Magnetic Fields, Strengths

Frequency	25 kV AC Railway	750 V DC Railway
Hz	A/m	A/m
Zero	-	100
50	20	-
150	7	-
250	3	-
300	-	2
350	2	-
450	2	-
550	1	-
600	-	1
650	1	-
750	1	-

4.2.4

Under short circuit conditions, higher currents will flow for short times and these will produce higher magnetic fields for the duration of the short circuit than those in Table 1. The maximum value of field at the boundaries under short circuit conditions should not exceed:

Code of Practice

GM/RC 1500

Issue One

Date December 1994

Page 6 of 9

Code of Practice for EMC Between the Railways and its Neighbourhood

a) 600 A/m for a 750 V DC railway

b) 100 A/m for a 25 kV AC railway

Note 2: On a 25 kV railway, the short circuit can contain harmonics, but for the purpose of this Code of Practice, it can be considered as a 50 Hz current, without harmonics.

4.2.5

The Controller should ensure that radio frequency (RF) field gradients, at the boundary planes, under normal operating conditions, do not exceed the values given in Table 2.

RF is internationally defined as those between 9kHz and 400GHz.

4.2.6

Table 2 - Permitted Field Strengths

Radiated Emission Limit expressed as the points on a log/log graph which are joined by straight lines.

Limit Peak Value		
Frequency MHz	25kV AC Railway	750V DC Railway MHz
	dB(μ A/m)	
0.009	60	45
0.15	35	25
0.15+	65	55
30 -	15	5
	dB(μ V/m)	
30+	100	80
1000	65	50

These values are based on the 80/80 confidence level rule as defined in EN 55013.

The + and - signs indicate the places where the International Special Committee for the Protection of Radio (CISPR) bandwidth changes, resulting in a change of limit value for the emission.

Note 3: The values in Table 2 are based on measured values obtained by CENELEC.

4.2.7

The electric field at the boundary planes should not exceed 1200 V/m peak.

Code of Practice for EMC Between the Railways and its Neighbourhood

Code of Practice

GM/RC 1500

Issue One

Date December 1994

Page 7 of 9

4.2.8

At the points of connection to the Electricity Supply Industry, the railway installation should meet limits agreed between the Controller and the electricity supplier.

4.2.9

The Controller should ensure, as far as is reasonably practicable, that no unacceptable interference is introduced on Telephone Operator lines as a result of operation of the railway.

4.2.10

In cases where another railway system is operating nearby the Operators of both railways should comply with mutually agreed limits relating to electromagnetic compatibility effects.

4.3 Immunity

4.3.1

In some areas of operation the railway installation may be subjected to RF emissions which have the following characteristics:

- a) *Applied Field strength 20 V/m (146 dB(μ V/m))*
- b) *Frequency Range 0.15MHz to 500 MHz, with on/off keying at 1Hz.*

Note 4: *In some areas of operation the railway may have to operate with high voltage overhead electricity cable running along or across the railway, and/or in an environment which includes RF radiation from broadcasting stations, airport, aircraft, military radar, and portable sources such as radio telephones which may generate much higher fields than those which the railway produces itself. Such strengths have been found, for example, near Short Wave public broadcasting transmitters.*

The installation may have to be immune to radio frequency fields below 0.15 MHz and above 500 MHz (e.g., cellular telephones have a field strength exceeding 10V/m), at the levels appropriate to a specific location.

In setting the radio frequency immunity limits it is recognised that it is impossible to test the railway installation to these limits. The Controller and the Operator should take all reasonable steps to ensure that the railway and trains are immune should this radio field be applied.

Code of Practice

GM/RC I 500

Issue One

Date December 1994

Page 8 of 9

Code of Practice for EMC Between the Railways and its Neighbourhood

5 Guidance Notes

5.1

It is recognised that non-railway apparatus may be required to operate in the space above or below the railway or within the 10 m lateral boundaries of the railway and the Controller should, when requested by a relevant authority, predict the magnetic fields in these spaces.

These predictions should include values for the fields produced by signalling track circuit currents if a rail carrying track circuit current is less than 3 m from the space which contains the non-railway apparatus.

5.2

Apparatus beyond the railway boundary will be immune to the RF emissions specified in Table 2 if the apparatus meets the immunity requirements of prEN 50082-2

Code of Practice for EMC Between the Railways and its Neighbourhood

Code of Practice

GM/RC1500

Issue One

Date December 1994

Page 9 of 9

References

EMC *Statutory Instrument 1992 No. 2372* **Electromagnetic Compatibility**
The Electromagnetic Compatibility Regulations 1992.

EN 55013 *Limits and methods of measurement of radio frequency characteristics of Sound
and Television broadcast Receivers and associated Equipment*